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What is a Perihilar Cholangiocarcinoma (Klatskin tumor) & how to do the diagnosis?

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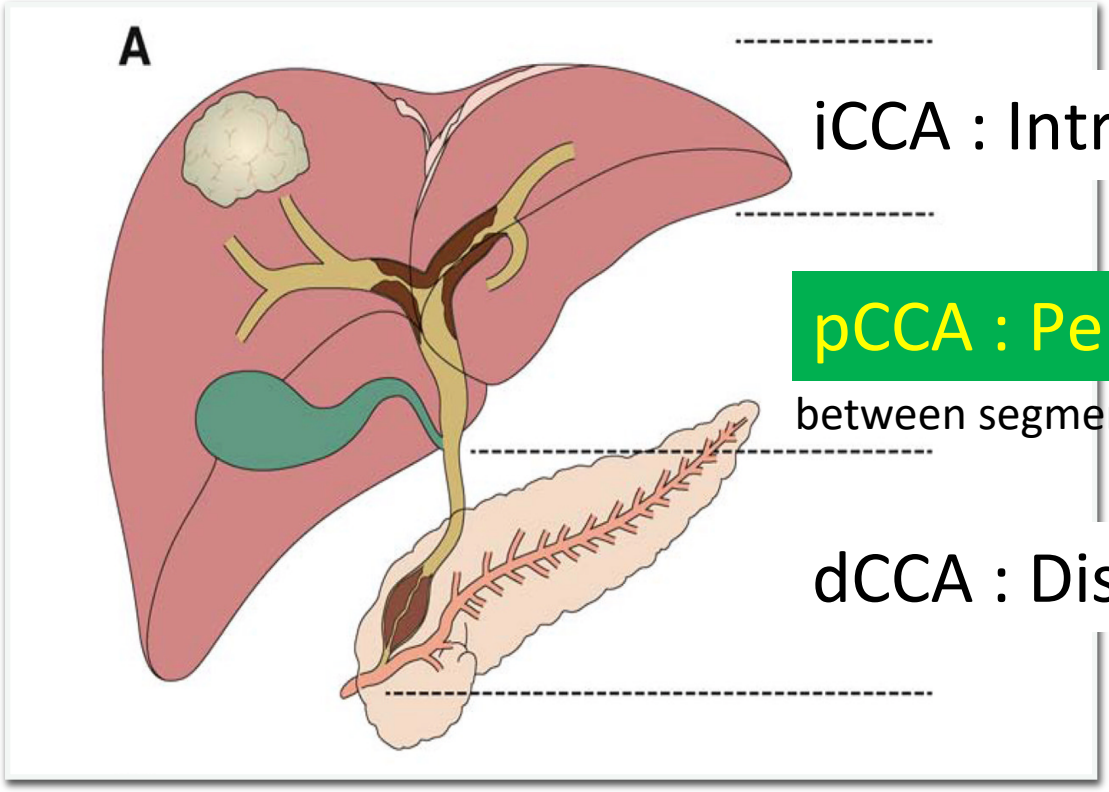
Erasme-ULB University Hospital – Brussels (Belgium)

HBPS – PGC 2019

Cholangiocarcinoma

Malignant tumor rising from bile duct epithelial cells



iCCA : Intrahepatic CCA — 10-20%

pCCA : Perihilar CCA — 50%

between segmental bile ducts and cystic duct

dCCA : Distal CCA — 30-40%

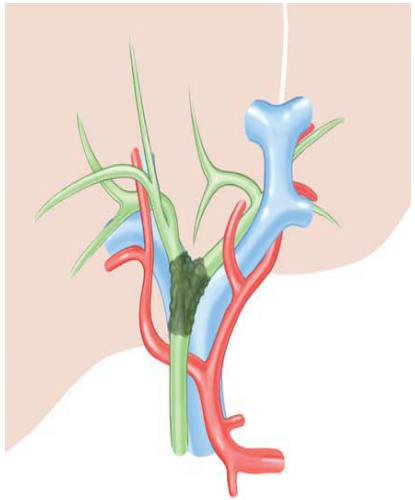
… terms of Klatskin tumor and Extrahepatic CCA should be avoided

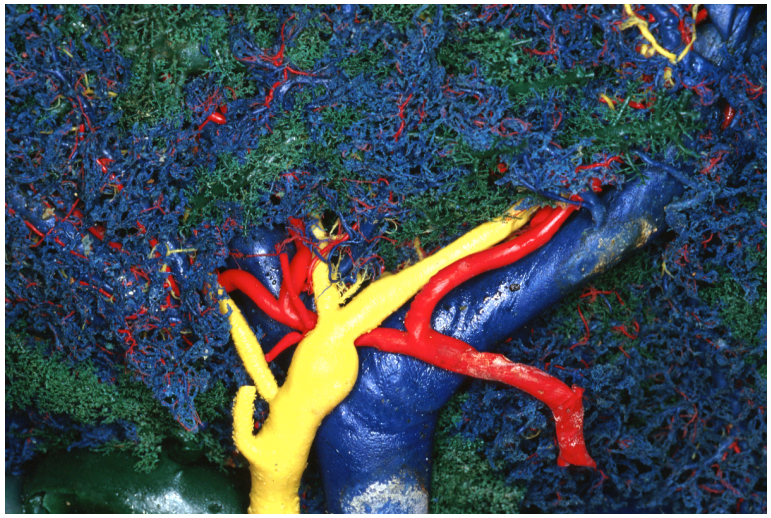
Rizvi et al. Gastroenterology 2013
Bridgewater et al. EASL guidelines: J Hepatol 2014

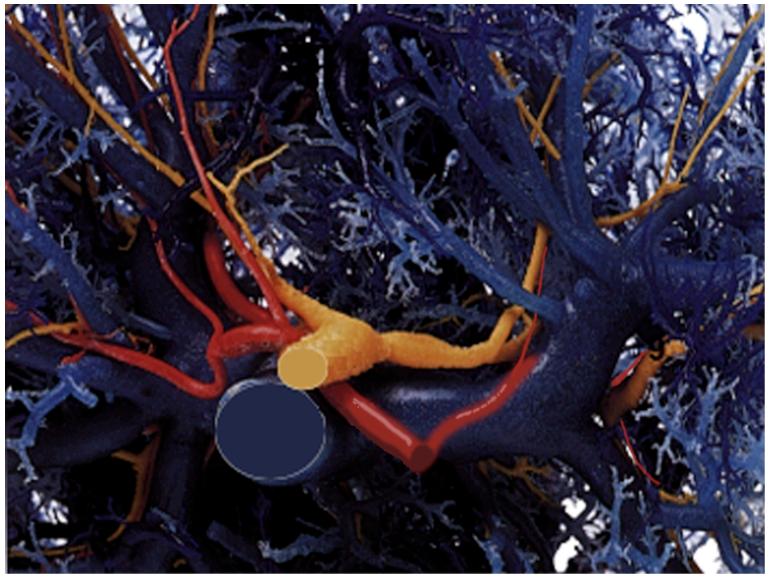
pCCA <3% of digestive cancers

Particular problem of pCCA

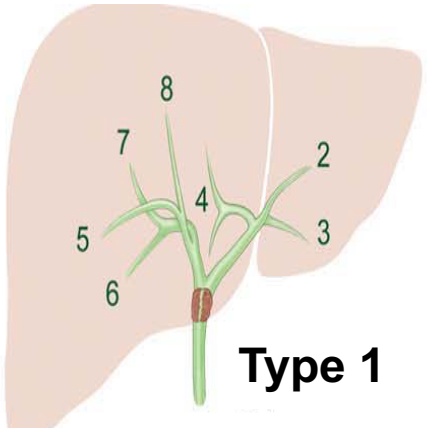
- Size
- Location
- Proximity with vessels
- Diagnosis



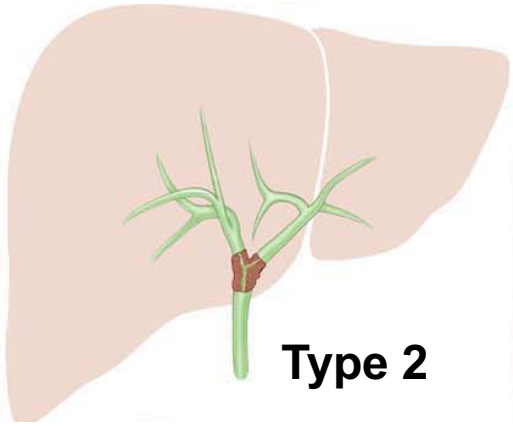




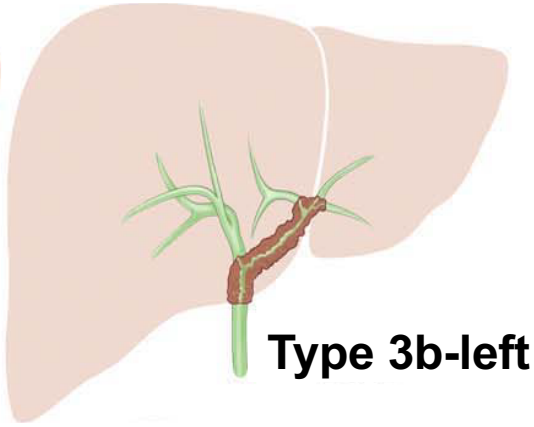
Classification of pCCA: Bismuth-Corlette



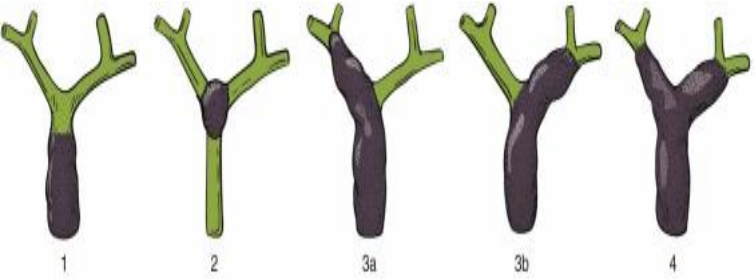
Type 1

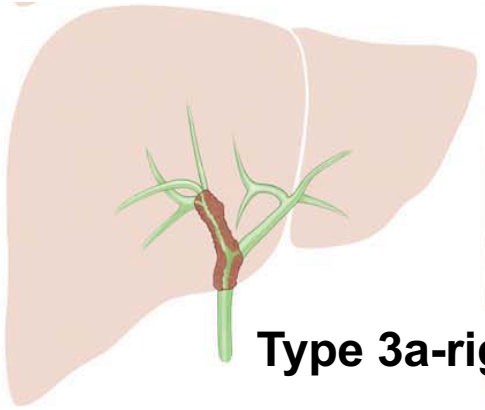


Type 2

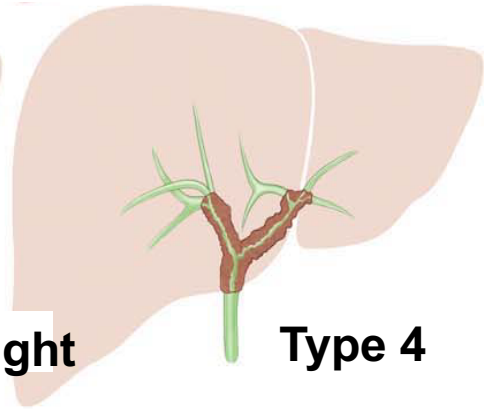


Type 3b-left





Type 3a-right



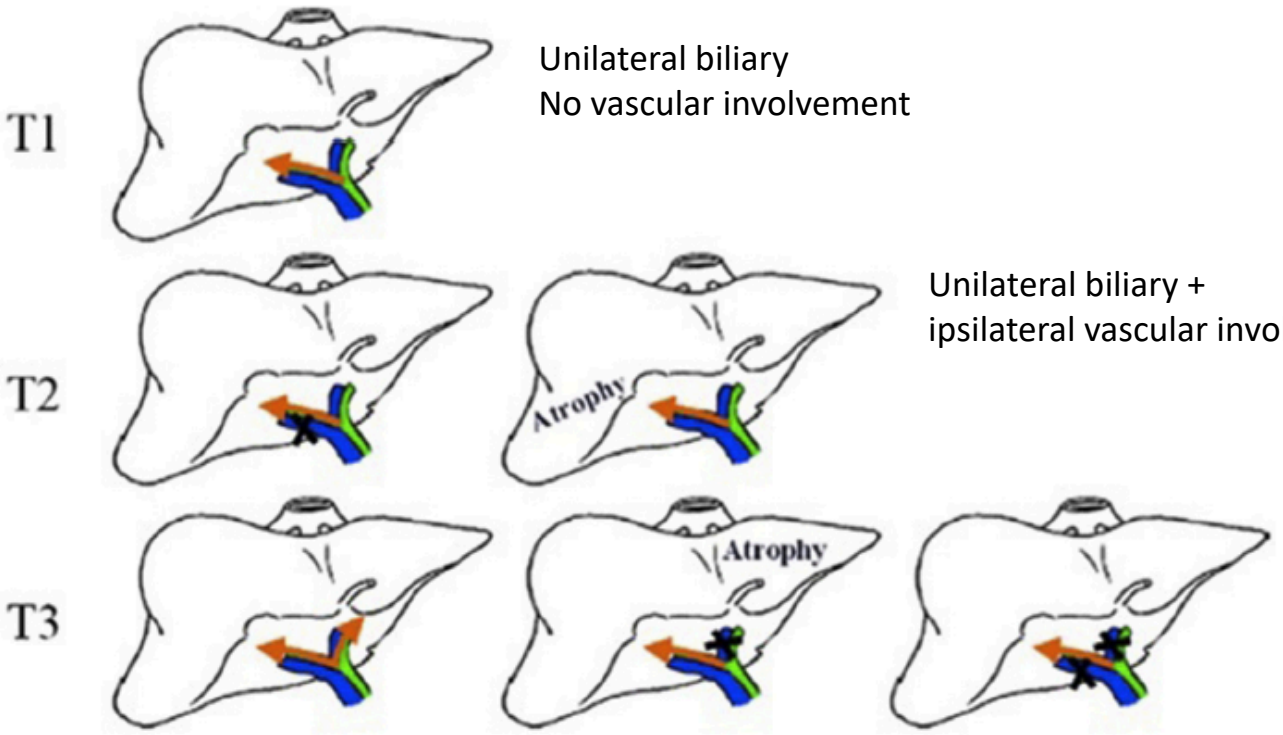
Type 4

Stratifies patients by longitudinal extension

- it ignores vascular involvement

Bismuth, Corlette et al. Surg Gynecol Obst 1975

Classification of pCCA: Jarnagin-Blumgart MSKCC System

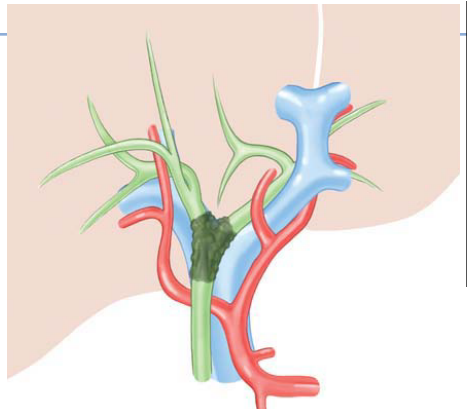


Stratifies patients by longitudinal and radial extension

- extension of tumour in to the vascular structures
- surrogate indicators of radial tumour extension (atrophy)

Jarnagin WR et al. Ann Surg 2001

Classification of pCCA: … B-HA-PV-V

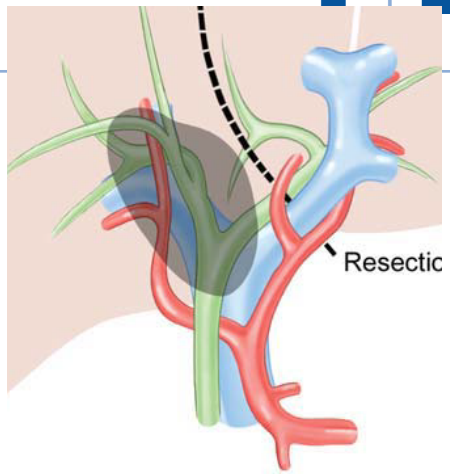


B2, PV0, HA0

New Staging System and a Registry for Perihilar Cholangiocarcinoma

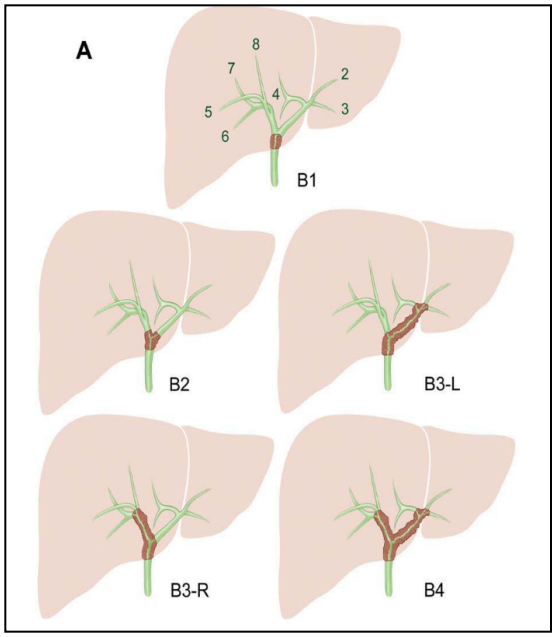
Michelle L. DeOliveira,[1] Richard D. Schulick,[2] Yuji Nimura,[3] Charles Rosen,[4] Gregory Gores,[5] Peter Neuhaus,[6] and Pierre-Alain Clavien[1]

De Oliviera et al. Hepatology 2011

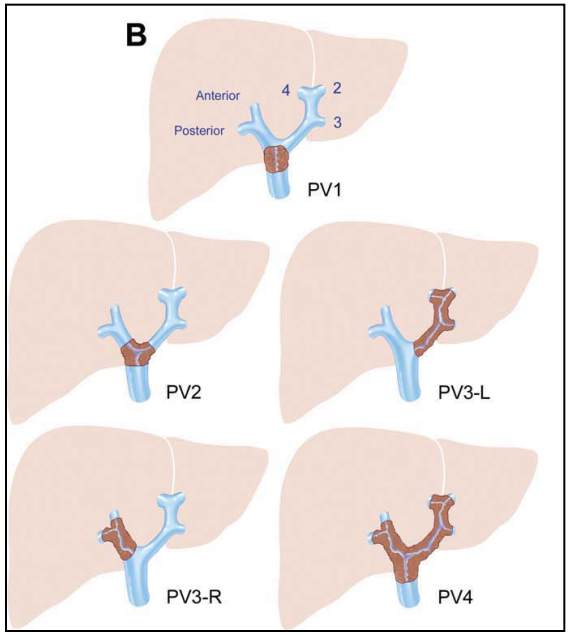


B3-R, PV3-R, HA3-R

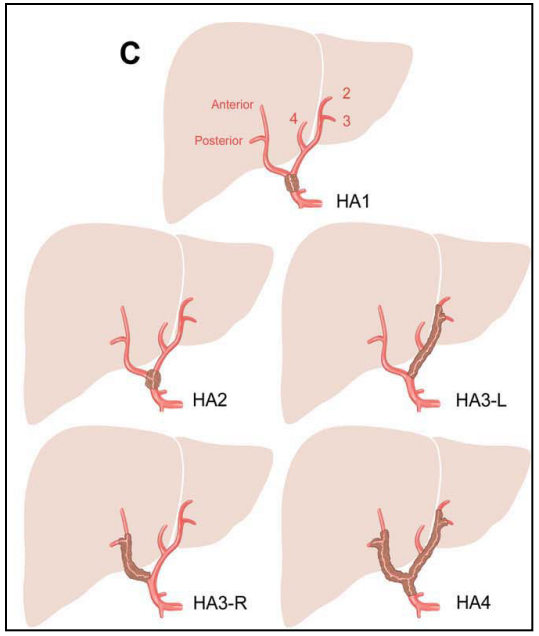
Longitudinal + Radial + Volume RL



Longitudinal Biliary extension



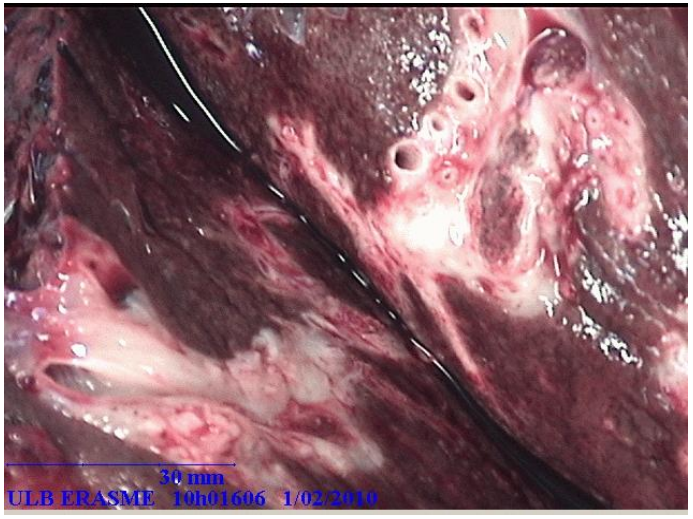
Radial Portal infiltration

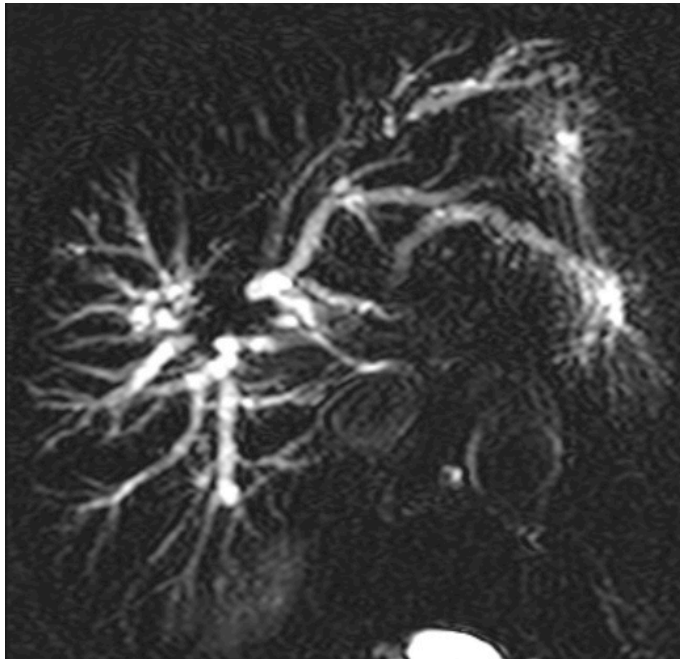


Radial Arterial infiltration

Main caveats

- Differential diagnosis
  - Fibrotic & infiltrative tumor
- Invasive diagnostic exams: consider surgical strategy





Differential diagnosis of Hilar stenosis

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BENIGN

- PSC – Primary Sclerosing Cholangitis
- BDI – Bile Duct Injury post cholecystectomy (traumatic or ischemic)
- Mirizzi Syndrome
- IgG4 mediated inflammatory stenosis
  - Hilar stenosis without visible mass
  - 50% combined pancreatic anomaly
  - IgG4 positif Dosage in 60%
  - Test treatment by corticosteroids 1 month:
    - Decrease of IgG4 and CA19.9 levels
- Portal Biliopathy (Biliary compression due to portal cavernoma)

MALIGNANT

- Intrahepatic Cholangiocarcinoma infiltrating the hilum
- Gallbladder cancer infiltrating the hilum
- Metastatic infiltration of other cancer or peritoneal carcinomatosis

Buc et al. HPB 2008
Hadjis et al. Br J Surg 1985

Differential diagnosis of Hilar stenosis

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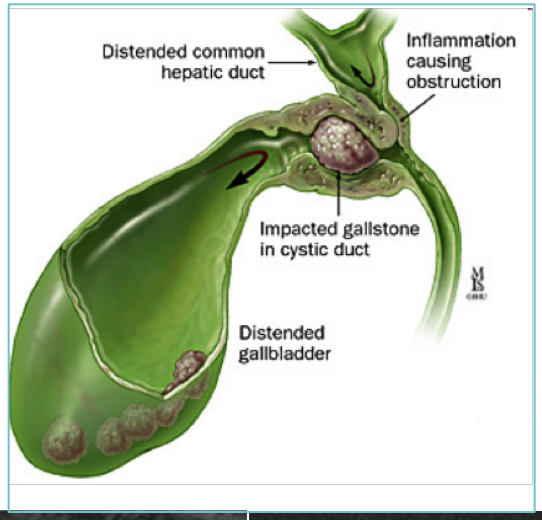
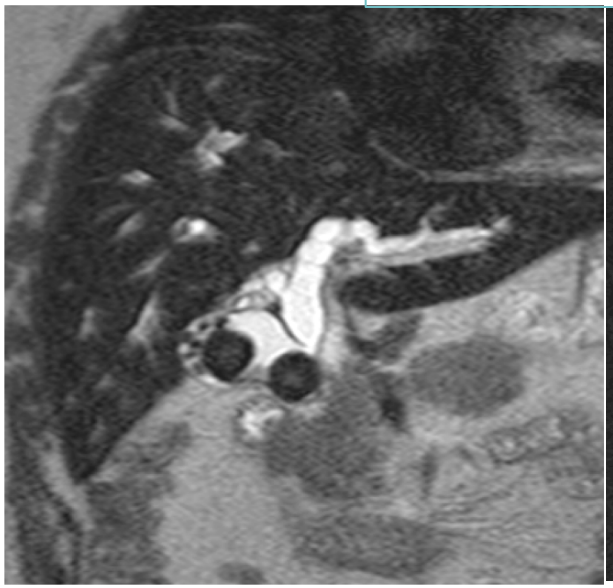
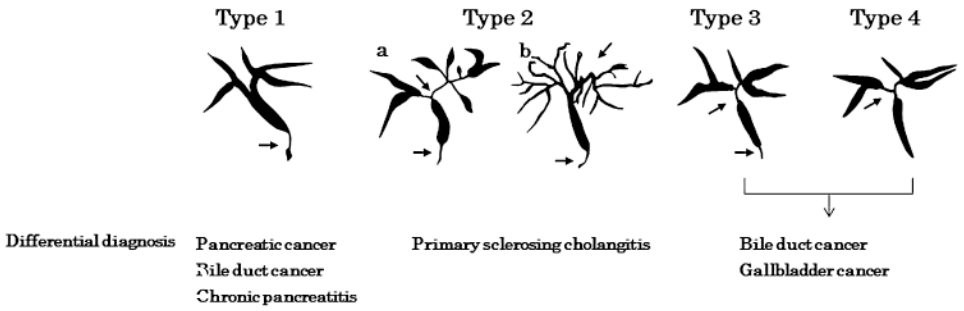
Traumatic BDI

Mirizzi Syndrome

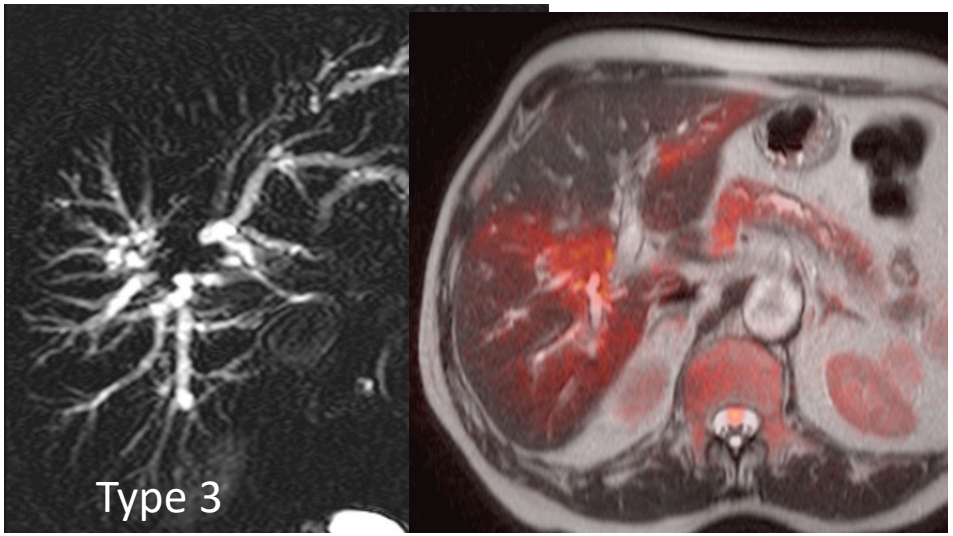


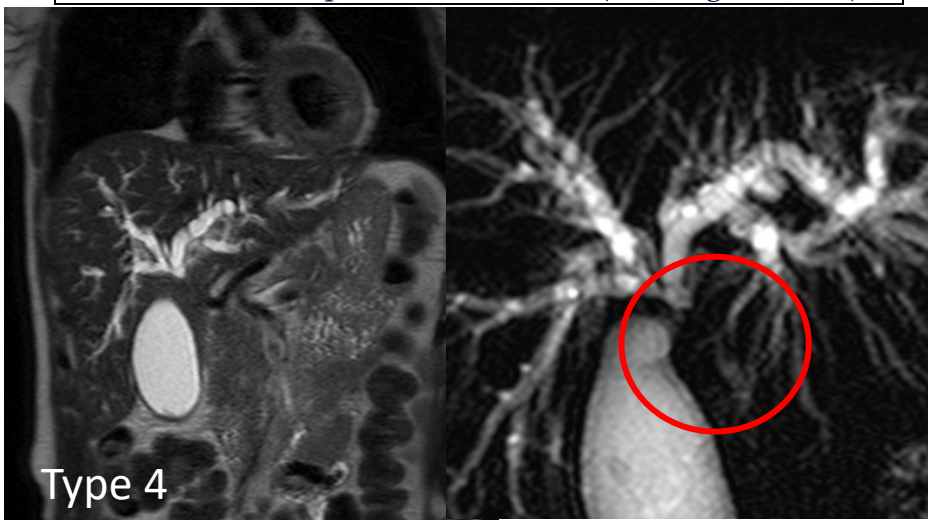


IgG4-related cholangitis: differential diagnosis



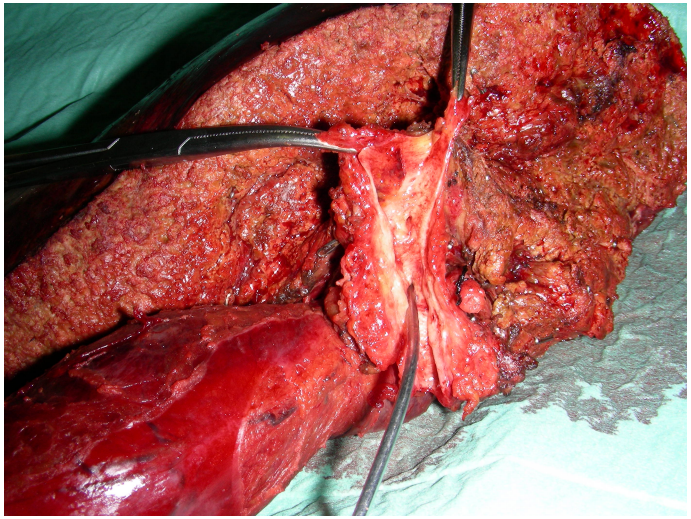
- IgG4-related disease:
  - pancreatitis (60%)
  - sialoadenitis (34%)
  - nephritis (23%)
  - dacryoadenitits (23%)
  - periaortitis (20%)
  - cholangitis (13%)
- M>F (4:1) middle aged or the elderly
- 80% have elevated serum IgG4 (> 135mg/dL)
- At imaging: may mimick other biliary disorders
- Lymphoplasmacytic infiltration, including abundant IgG4-bearing plasma cells in the bile duct lesions
- Favorable response to treatment (stenting, steroids)

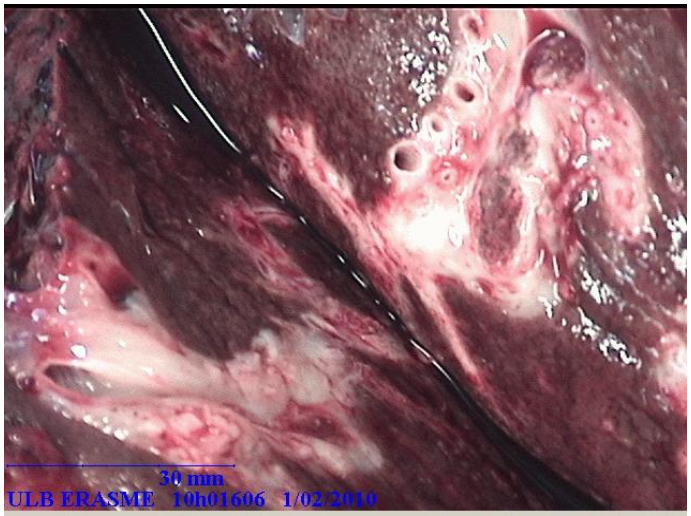




Almost certain diagnosis of pCCA

- Jaundice or biological cholestasis
- Regular-shaped IHBD
- Small but visible hilar mass
- No previous cholecystectomy
- No portal cavernoma
- Normal pancreas





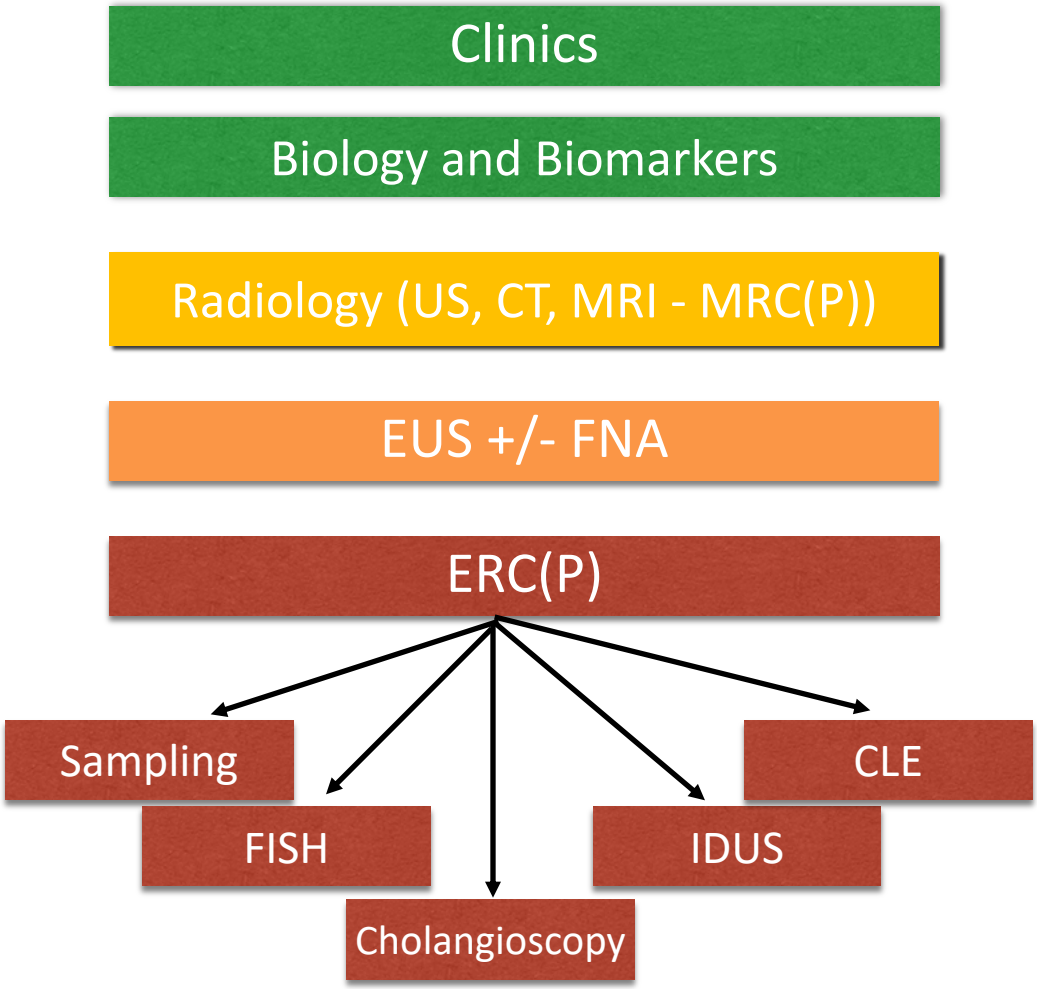
Differential diagnostic approach in pCCA

Hilar Cholangiocarcinoma (80%)

VS

- Benign stricture (post surgical)
- Lithiasis
- IgG4 cholangiopathy
- PSC
- Other tumor (benign or malignant)

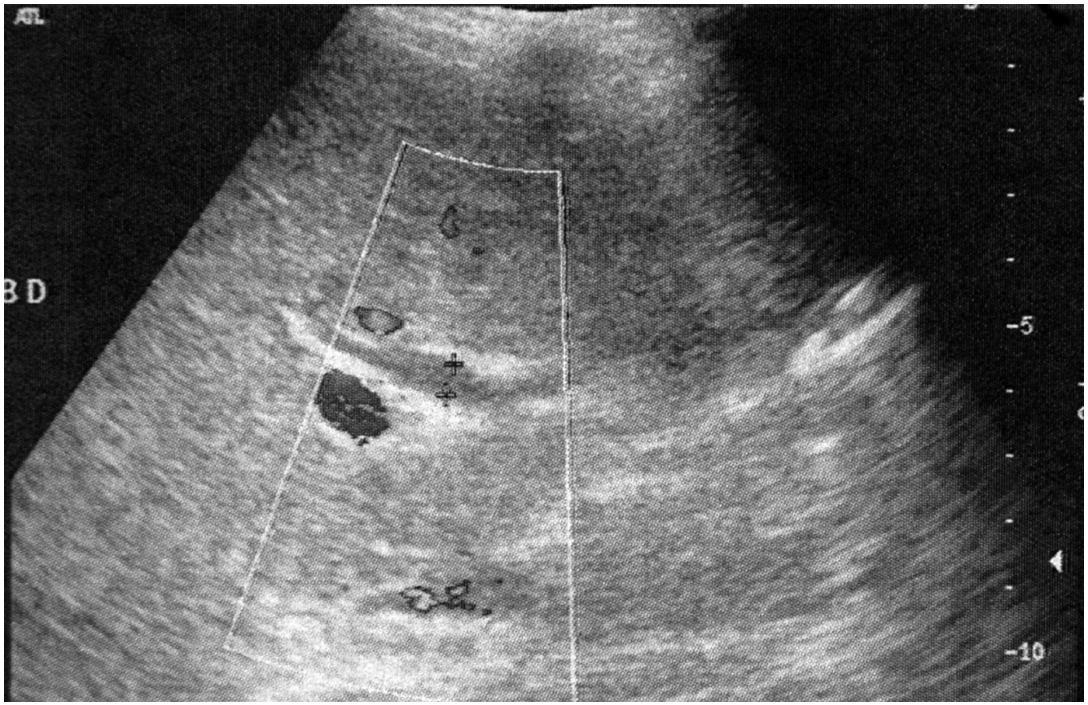
Are. Surgery. 2006



Clinical symptoms

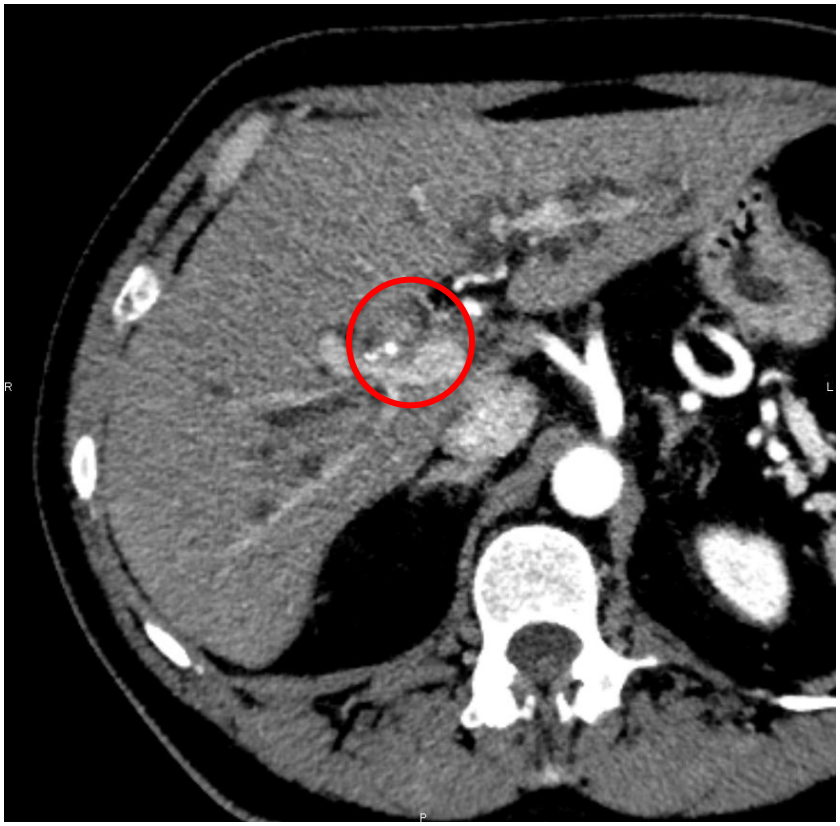
- Biological cholestasis (may be isolated)
- Jaundice (70%)
- Dilatation of IHBD
- Non-dilated gallbladder
- nonspecific symptoms including abdominal discomfort, cachexia, weight loss, and malaise typically consistent with biliary obstruction presenting with jaundice and less commonly cholangitis

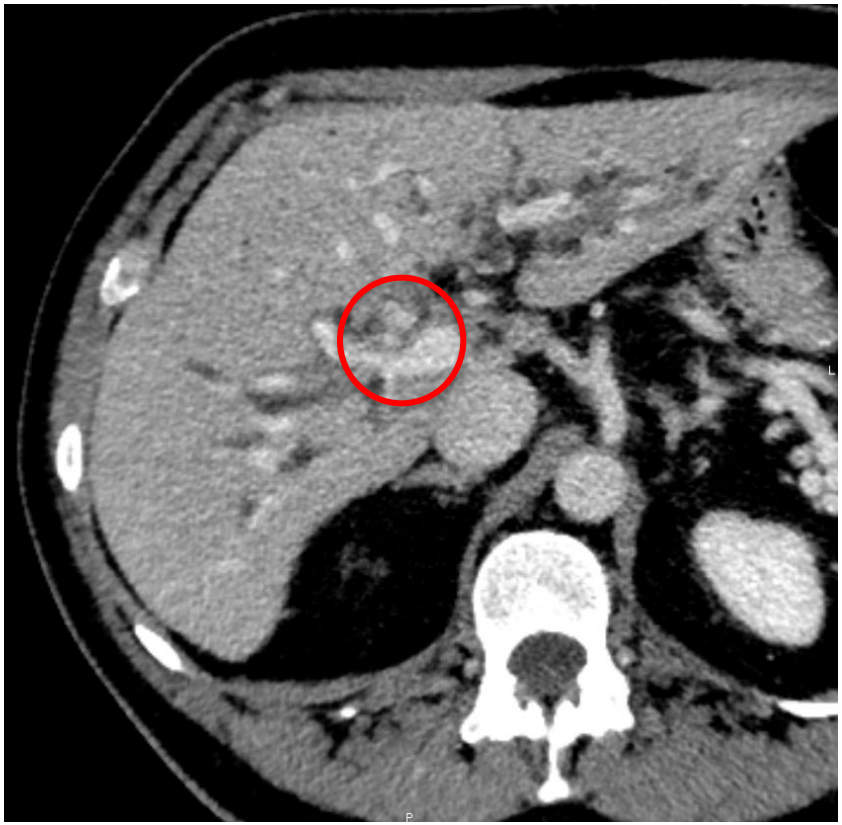
Liver Ultrasound



- Dilatation of Intra-Hepatic Bile Ducts
- Non distended gallbladder
- Hilar mass ?

CT

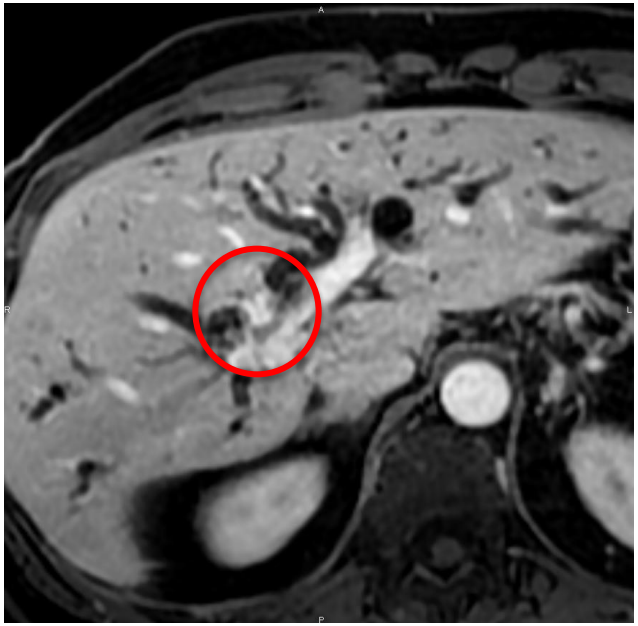


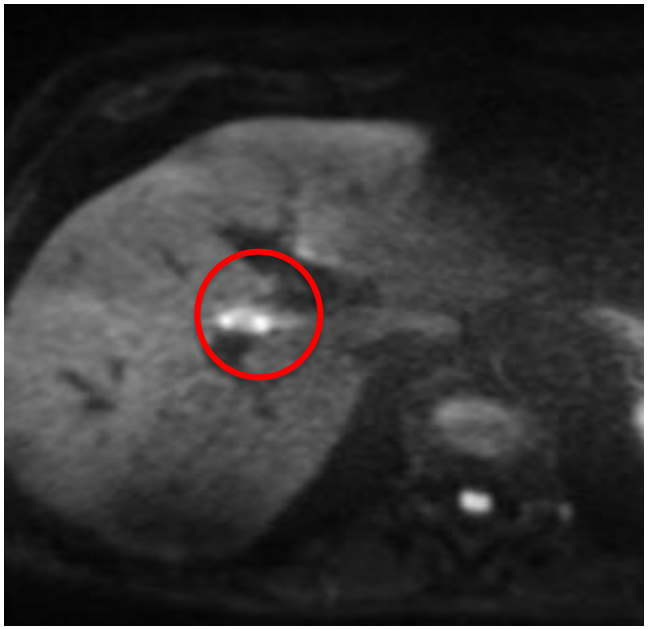


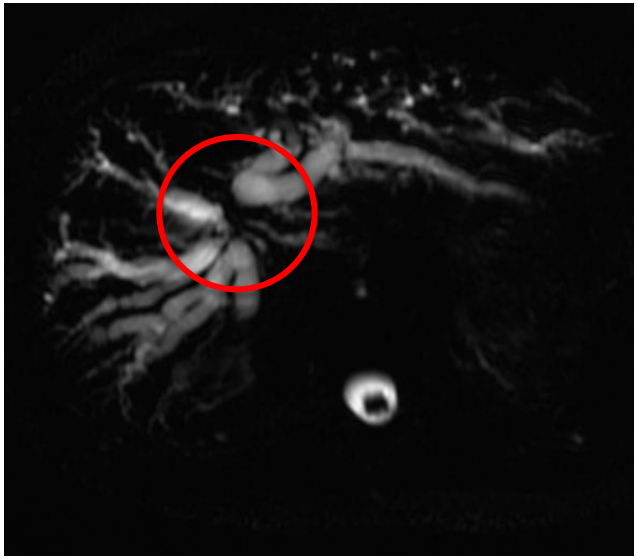
- Dilatation of Intra-Hepatic Bile Ducts
- Non distended gallbladder
- Hilar mass
- Vascular behaviour of mass

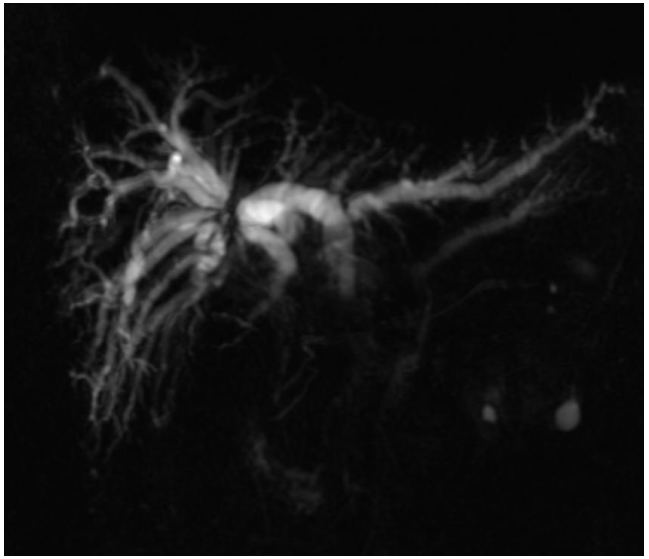
- Portal vein invasion ?
- Arterial branch invasion?
- Hilar & coeliac adenopathies ?
- Peritoneal seeding?

MRI





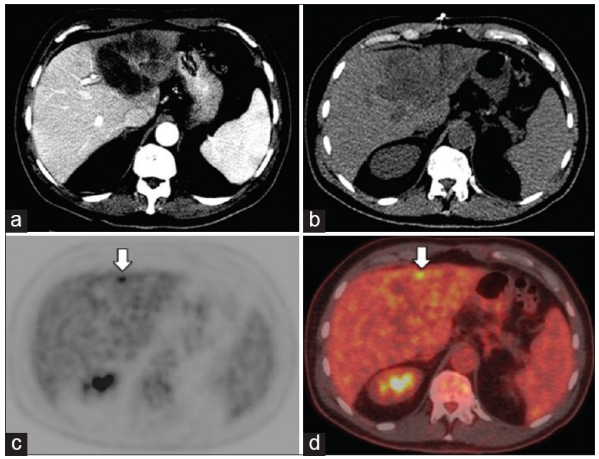


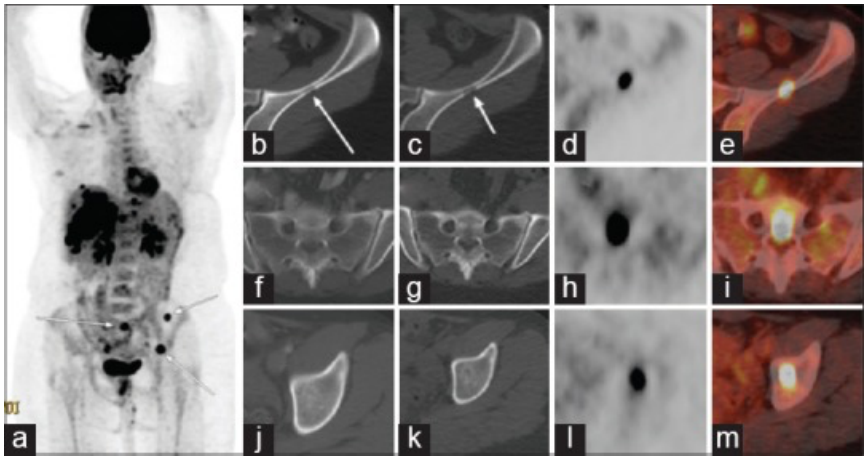


PET-scan

	FDG PET/CT	CECT	P
Sensitivity (%)	96.5 (91.61,98.71)	62.2 (53.72, 70.09)	<0.0001
Specificity (%)	55.5 (22.65, 84.65)	66.7 (30.92, 90.96))	NS
PPV (%)	97.2 (92.50, 99.09)	96.7 (90.09, 99.15)	NS
NPV (%)	50.0 (23.66, 76.34)	10.0 (4.13, 21.17)	0.0064
Accuracy (%)	94.1 (88.72, 97.08)	62.5 (54.25, 70.11)	<0.0001

FDG PET/CT: Fluoro-D-glucose positron emission tomography/computer tomography; CECT: Contrast enhanced computed tomography; PPV: Positive predictive value, NPV: Negative predictive value, values in parenthesis are 95% confidence intervals



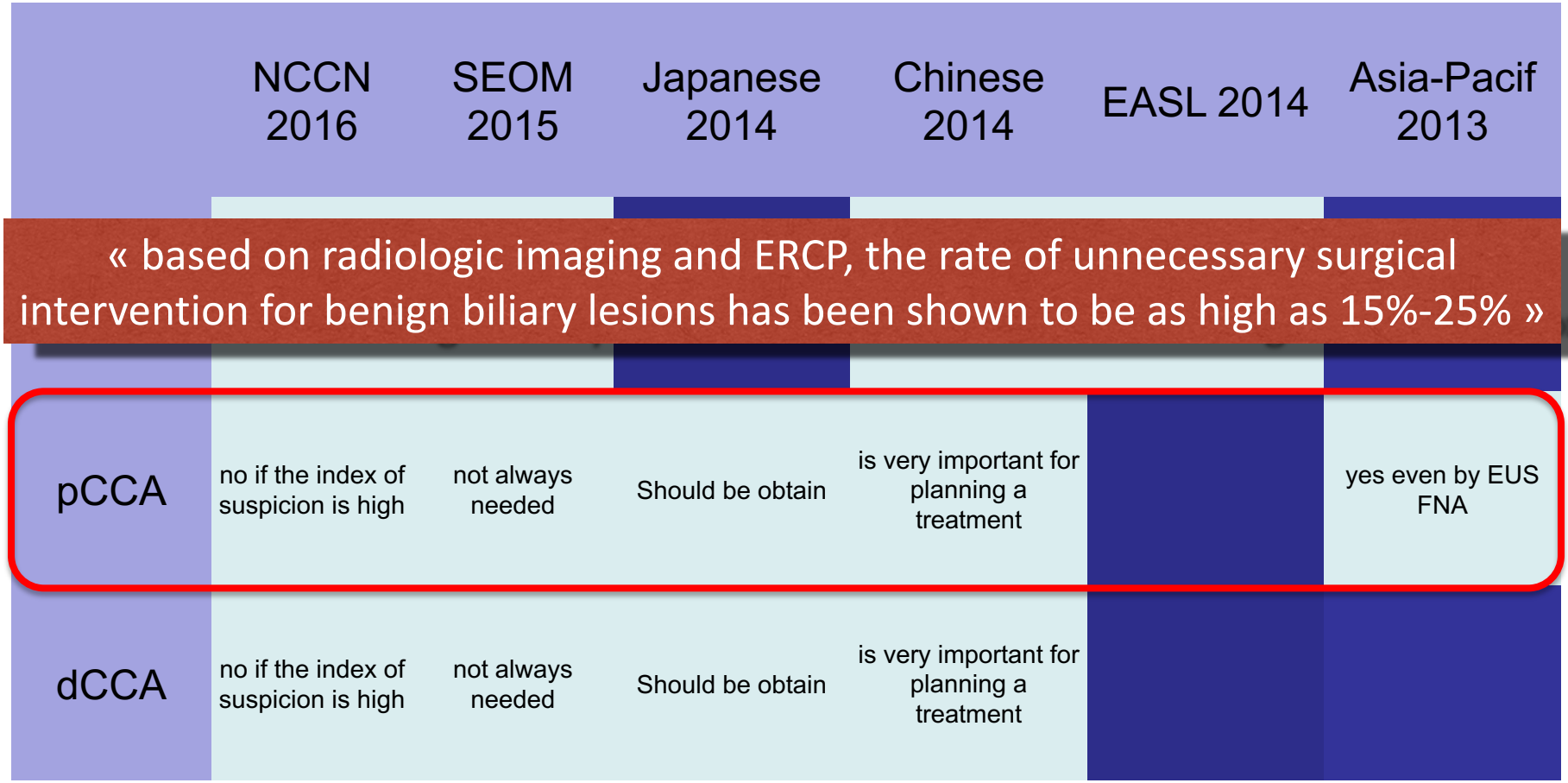


Pre-operative Histological proof ?

	NCCN 2016	SEOM 2015	Japanese 2014	Chinese 2014	EASL 2014	Asia-Pacif 2013
iCCA	no for non-cirrhotic	not always needed		is very important for planning a treatment	no for non-cirrhotic	
pCCA	no if the index of suspicion is high	not always needed	Should be obtain	is very important for planning a treatment		yes even by EUS FNA
dCCA	no if the index of suspicion is high	not always needed	Should be obtain	is very important for planning a treatment		

Benson 2009, Benavides 2015, Miyazaki 2015, Xiao-ping 2014, Bridgewater 2014, Rerknimitr 2013

Pre-operative Histological proof ?



Benson 2009, Benavides 2015, Miyazaki 2015, Xiao-ping 2014, Bridgewater 2014, Rerknimitr 2013

Do we need to have histology?

- NO ERCP before analysis of imaging work-up by specialized HB-surgical team !!!

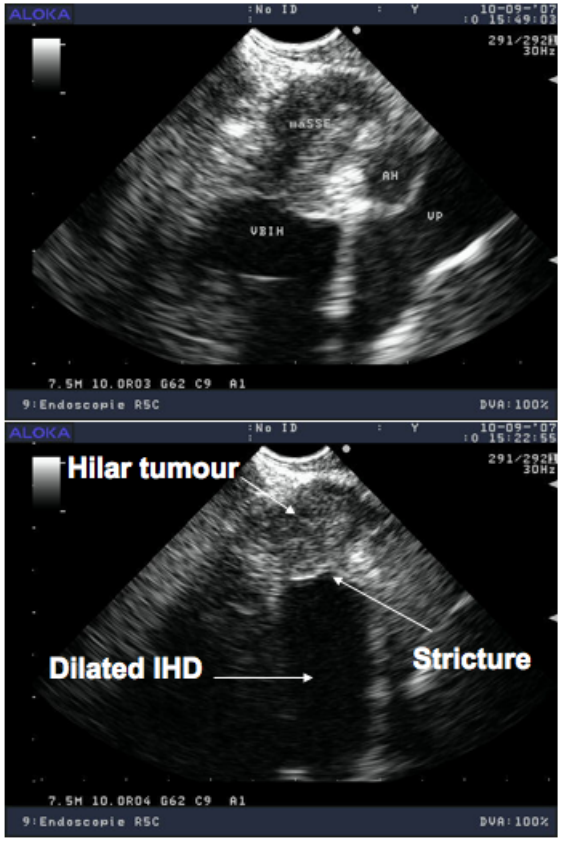
Endoscopic US + FNA

CHU Saint-Pierre UMC Sint-Pierre | Hôpital Erasme | ULB

Role of EUS for preoperative evaluation of cholangiocarcinoma: a large single-center experience (CME)

Mehdi Mohamadnejad, MD, John M. DeWitt, MD, Stuart Sherman, MD, Julia K. LeBlanc, MD, Henry A. Pitt, MD, Michael G. House, MD, Kelly J. Jones, MD, Evan L. Fogel, MD, Lee McHenry, MD, James L. Watkins, MD, Gregory A. Cote, MD, Glen A. Lehman, MD, Mohammad A. Al-Haddad, MD

Indianapolis, Indiana, USA; Tehran, Iran



- High level of tumor detection *
  - compare to CT and MRI
- EUS **increase the detection of unresectable lesion** *
  - vascular invasion
  - liver M+
  - Coeliac and distant LN
- Sensitivity of EUS-FNA is significantly higher in distal than in proximal CCA.

*Mohamadnejad et al. GIE 2011 **
*Weilert et al. GIE 2014 ***
*Moura et al. Endoscopy International Open 2018 ***

Endoscopic US + FNA

CHU Saint-Pierre UMC Sint-Pieter | Hôpital Erasme | ULB

ORIGINAL ARTICLE

Trans-peritoneal fine needle aspiration biopsy of hilar cholangiocarcinoma is associated with disease dissemination

Julie K. Heimbach, William Sanchez, Charles B. Rosen & Gregory J. Gores

William J von Liebig Transplant Center, Mayo Clinic College of Medicine, Rochester, MN, USA

« Transperitoneal biopsy of hilar cholangiocarcinoma is associated with a higher rate of peritoneal metastases, and it **should not be performed if a curative approach is available** »

Table 2 Incidence of peritoneal metastasis in patients who underwent a diagnostic transperitoneal fine needle aspiration (FNA) biopsy of hilar cholangiocarcinoma (CCA)

	No biopsy	Positive transperitoneal biopsy	Negative transperitoneal biopsy
Peritoneal metastasis at staging	14/175 (8%)	5/6 (83%)[a]	0/9
	14/175 (8%)	**5/15 (33%)[b]**	

[a]$P = 0.0001$ using Fisher's exact test.

Heimbach et al. HPB 2011

Endoscopic US + FNA

- YES for local US evaluation of pCCA
- YES for Lymph-nodes biopsies
- NO for direct hilar mass biopsy

ERCP : brush and biopsy

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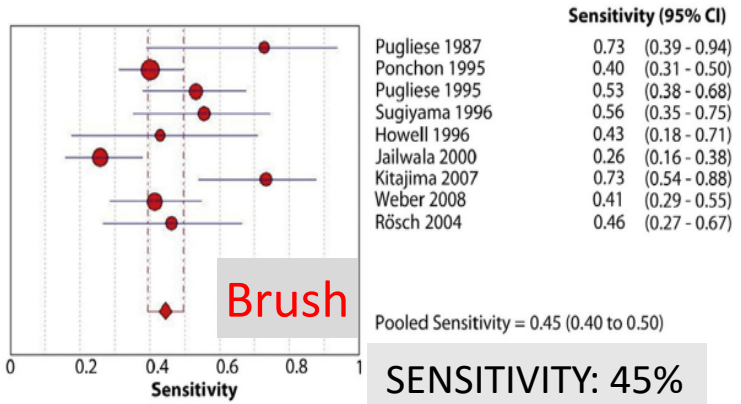
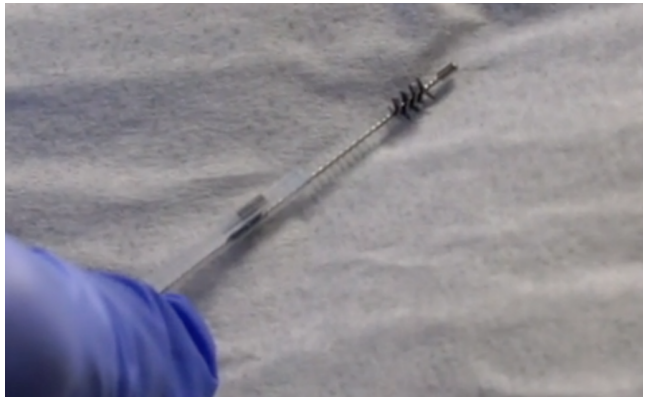
Comparative effectiveness of biliary brush cytology and intraductal biopsy for detection of malignant biliary strictures: a systematic review and meta-analysis

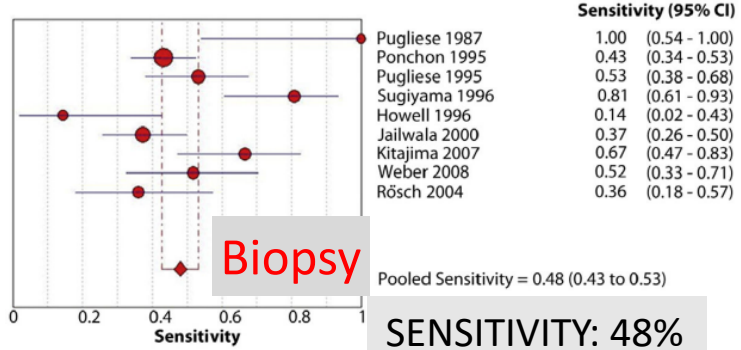
Udayakumar Navaneethan, MD[1,2], Basile Njei, MD, MPH[3], Vennisvasanth Lourdusamy, MD[1],

Navaneethan GIE 2015



SENSITIVITY: 45%





SENSITIVITY: 48%

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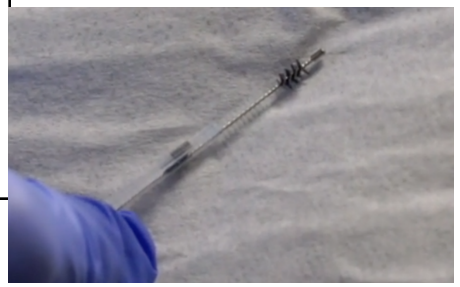
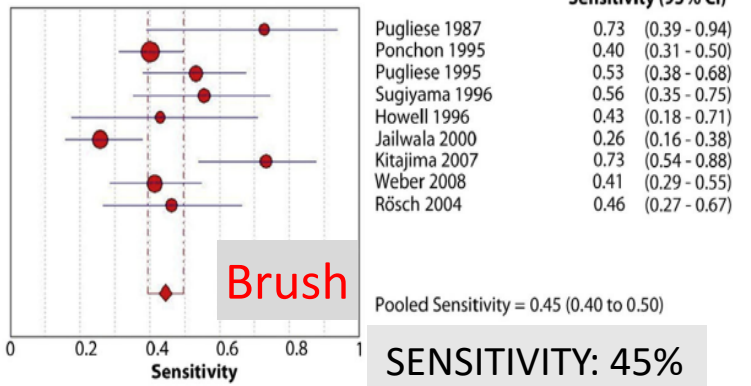
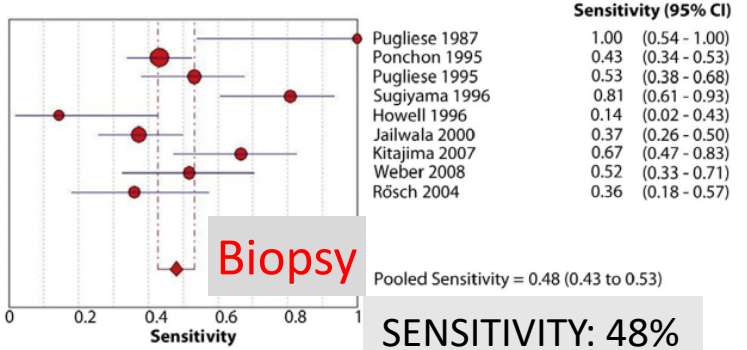
Navaneethan GIE 2015

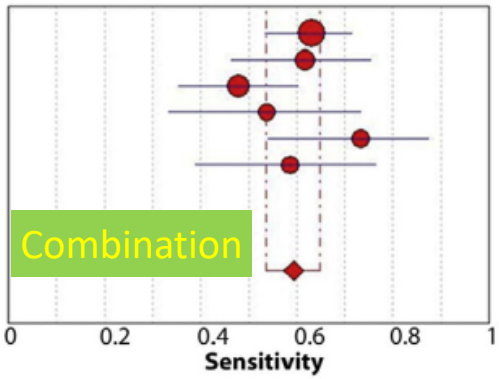




SENSITIVITY: 45%

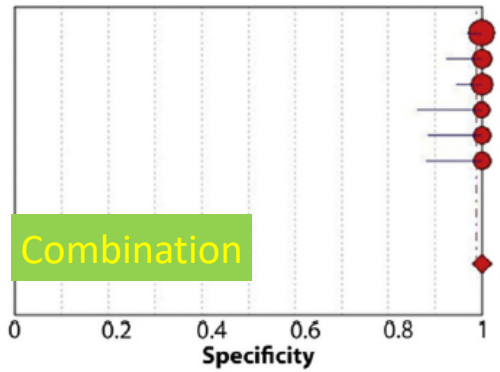


SENSITIVITY: 48%



SENSITIVITY: 59%

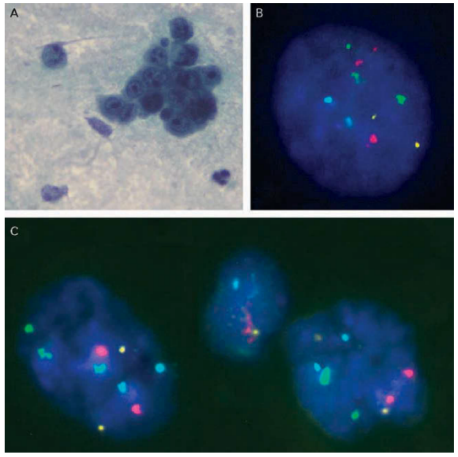
SPECIFICITY: 100%



ERCP : brush and FISH

CHU Saint-Pierre UMC Sint-Pieter | Hôpital Erasme | ULB

FISH : Fluorescence *in situ* hybridization

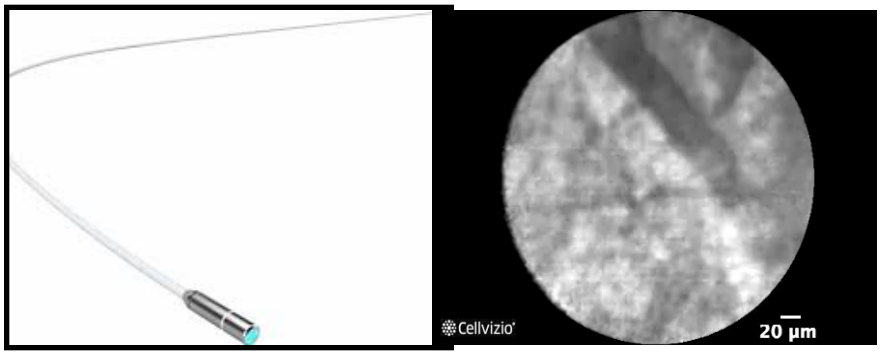


SENSITIVITY: 50-80%
SPECIFICITY: … up to 100%

Spyglass

Intraductal US

Confocal Laser Endomicroscopy



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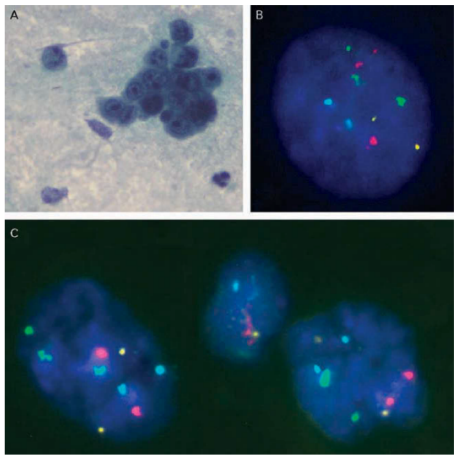
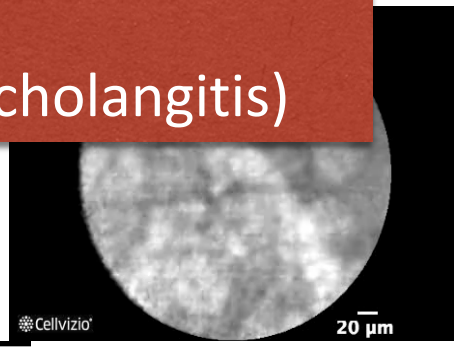


Spyglass

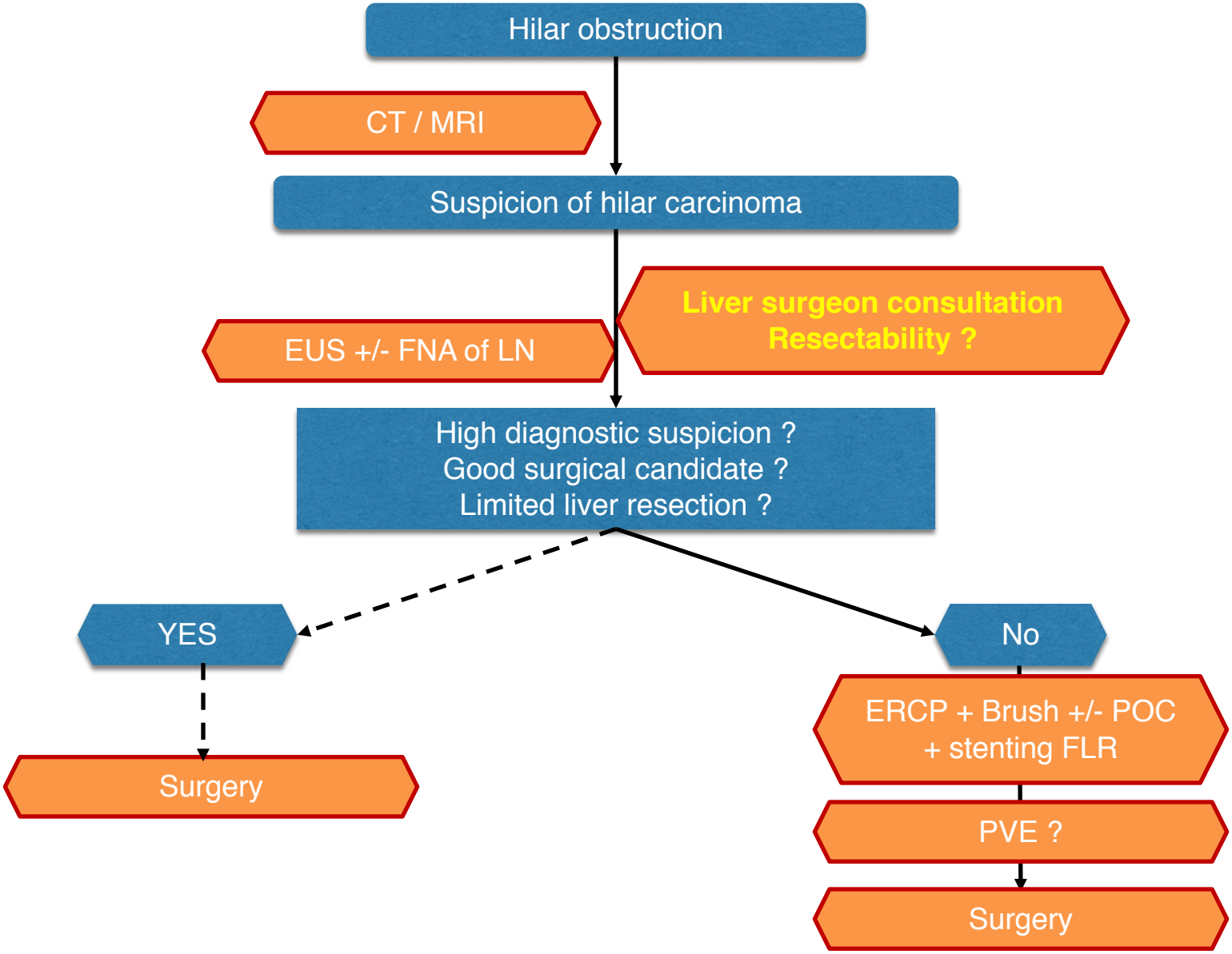
SENSITIVITY: 50-80%
SPECIFICITY: … up to 100%

Drawbacks :

- Expert center / hands
- Cost +++
- Complication rate around 7.1% (most common cholangitis)



Work-up for pCCA diagnosis



Take home message

Hôpital Erasme

ULB

- Hepato-Biliary Surgeon has to determine resectability at initial diagnosis : he decides on necessity of invasive exams
- Histological proof not indispensable to go to surgery
- Transperitoneal puncture of the lesion to be avoided!!!
- Exclude metastatic sites: MRI / AngioCT / EUS+/-FNA ADP / PET / CT Thorax
- Biliary drainage often needed after ERCP : type of drainage depending on established surgical strategy
  - Future liver remnant drainage